Date: 28 April 2026

From: S Veeraraju 48, Rehoboth, Evergreen Avenue, Shanti Medu, Billichi, Coimbatore – 641 019

To, The Commissioner of Police (Traffic) Coimbatore City Traffic Police Race Course Road, Coimbatore 641 018, Tamil Nadu

Subject: Academic Research Collaboration Request: AI Traffic Signal Fairness Study, UCL and Microsoft

---

Respected Sir / Madam,

I am writing this letter on behalf of my son, **Ebenezer Veeraraju**, who is a postgraduate student at **University College London (UCL)**, UK, completing his MSc dissertation on AI-driven traffic signal control. His thesis project is jointly supervised by Dr Akin Delibasi (UCL) and Mr Lee Stott (Microsoft UK).

He is a native of Coimbatore. Having had the opportunity to pursue postgraduate studies at UCL, he felt strongly that this research should benefit India directly rather than remain confined to a foreign country. The fieldwork is planned in Coimbatore from **25 May to 25 August 2026**.

What this project is

- Traditional traffic signals run on fixed timers regardless of actual traffic. AI-controlled signals read live queue lengths and adjust in real time.
- That AI systems can reduce overall waiting has already been shown. What has never been formally measured is whether they do so fairly.
- Does a busy commercial road get all the benefit while a residential street always waits? Do pedestrians lose out so vehicles can move faster?
- This research builds the first audit framework to answer these questions, using a computer simulation of a real city road network.
- Nothing will be installed or deployed on any actual junction.

What we are asking from you

Request	Details
Hourly vehicle flow counts	Total vehicles per road arm, per hour, at the signalised junctions along **Avinashi Road (NH-544), from Anna Silai (Anna Statue Junction) to Chinniyampalayam Junction** (approximately 11.6 km, covering all the junctions in that stretch), for the period **25 May 2026 to 25 August 2026**
Signal phase timings	Current green, amber and red durations at those junctions
A letter of support	One page on official letterhead confirming awareness of the research (draft provided separately)

No images. No CCTV. No licence plates. No personal data of any kind.

What effort is required from your side

- No exports or technical work required from your staff.

- My son would visit the control room personally as an observer and record the relevant figures manually from your existing screens.
- Alternatively, a brief meeting with a technical officer or a printed summary works equally well.
- We will work entirely around whatever is convenient for your office.

What Coimbatore gets

- A full AI equity audit report for the Avinashi Road corridor at no cost to the city
- Named acknowledgement in the UCL dissertation and any subsequent academic publications
- Recognition as the first Indian Smart City assessed against a formal AI traffic fairness standard
- Feature in a Microsoft technical blog article on this research, positioning Coimbatore as part of a UCL and Microsoft collaboration

---

Yours sincerely,

S Veeraraju (Father of Ebenezer Veeraraju, MSc Systems Engineering for IoT, University College London) 48, Rehoboth, Evergreen Avenue, Shanti Medu, Billichi, Coimbatore – 641 019

Research contact: Ebenezer Veeraraju ebenezer.veeraraju.25@ucl.ac.uk Supervisors: Dr Akin Delibasi, UCL and Lee Stott, Microsoft UK

(Technical details and data specifications are in the Appendix below.)

Appendix: For Further Reference

The AI System

The traffic controller uses Phi-4-mini, an open-source small language model built by Microsoft, running offline on a standard laptop-sized computer at each junction. It reads queue lengths, decides the next signal phase in under a millisecond and writes a structured reasoning record to a local blockchain ledger. No cloud connection is needed and there is no central server.

The Equity Audit

A formal audit runs every three months and measures the following:

- Gini coefficient: are waiting times distributed equally across all junction corridors?
- Disparate Impact Ratio: are some areas of the city consistently disadvantaged?
- Pedestrian parity: are pedestrian crossing phases proportionate to vehicle phases?

Each metric has a published pass or fail threshold. The audit produces a certificate that the city authority retains.

Why Coimbatore and Avinashi Road

Coimbatore is one of India's 100 Smart Cities under the Union Government's Smart Cities Mission. Traffic data for Coimbatore is already listed on the Government of India's Smart Cities Open Data Portal (smartcities.data.gov.in), confirming that the city's traffic infrastructure already supports structured data collection.

Avinashi Road (NH-544) is ideal for this research as it spans approximately 11.6 km from Anna Silai (Anna Statue Junction) to Chinniyampalayam Junction, passing through diverse land uses — dense city-centre activity near Anna Silai, busy commercial zones through the mid-corridor, and semi-urban areas near Chinniyampalayam. This gives the study a genuine range of traffic densities across all signalised junctions within a single contiguous corridor.

A comparable academic and police collaboration has previously been done in India. The Bengaluru City Traffic Police partnered with the Indian Institute of Science (IISc) to release the UVH-26 traffic dataset, which has since been cited in over 40 international research papers. We are seeking a similar spirit of cooperation, focused on fairness auditing rather than image data.

Data Protection

The aggregate vehicle counts requested are not personal data under India's Digital Personal Data Protection Act 2023 (DPDPA, Section 2(t)). No individual can be identified from hourly aggregate flow figures, and no data protection obligations arise on the Traffic Police from this collaboration.

Timeline

Milestone	Date
Initial data request	April 2026
Fieldwork begins in Coimbatore	25 May 2026
Data collection period (Avinashi Road, NH-544)	25 May 2026 to 25 August 2026
Dissertation submission	September 2026
Audit report delivered to Traffic Police	September 2026